

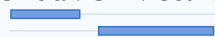
NOVA | Inference flow

Audio + query



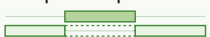
Generate candidates

Official / SFT / SetPO



Verify evidence*

Keep / Drop scores



Refine boundaries

Radius + DP + gate



Intervals

Query-aligned

*Routing follows the evaluation configuration.

RelTwin | Training detail

Paired examples

A: dog B: rooster



$W_+ : A \rightarrow B$ $W_- : B \rightarrow A$

$q_+ : A$ followed by B

$q_- : B$ followed by A

Same audio,
different correct windows.

Training objective

$$\mathcal{L} = \mathcal{L}_{\text{seq}} + \lambda \mathcal{L}_{\text{cand}} + \beta \mathcal{L}_{\text{replay}}$$

Likelihood scoring

Shared model

Frozen backbone

Trainable LoRA

Answer competition

$y(W_+)$

$y(W_-)$

q_+

q_-

$s_{++} \uparrow$

$s_{+-} \downarrow$

$s_{-+} \downarrow$

$s_{--} \uparrow$

s : mean token
log-likelihood

$\mathcal{L}_{\text{cand}}$: row-wise cross-entropy

Matched $\uparrow$

Swapped $\downarrow$

Localization example

Same audio; inverse queries

$q_+ : A \rightarrow B$ IoU

Target

SFT 0.00

RelTwin 0.69

$q_- : B \rightarrow A$ IoU

Target

SFT 0.98

RelTwin 0.98

Time (s) 0 10 20 27